

Bilal Shihab

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PROFESSIONAL SUMMARY

Biomedical Engineering undergraduate specializing in machine learning for clinical and biological data. Experience training and evaluating deep learning models across neural, physiological, and language modalities, from research prototypes to deployed products. Particularly focused on efficient LLM inference through quantization, KV-cache reuse, and memory-mapped deployment on constrained hardware.

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN

Austin, TX

Bachelor of Science in Biomedical Engineering, Computational Track

May 2027

- **Relevant Coursework:** Real-Time Digital Signal Processing, Statistics, Differential Equations & Linear Algebra, Numerical Methods, Intro to Computational Engineering Design, Systems & Signals, Biomedical Instrumentation, Embedded Systems

SKILLS

- **Languages:** Python, MATLAB, C++, C, ARMv6-M Assembly, R
- **ML & Inference:** PyTorch, llama.cpp, quantization, ONNX, ExecuTorch, TensorFlow Lite, Metal GPU inference, Optuna, Captum
- **Infrastructure & Embedded:** Docker, Google Cloud Platform, Git, Raspberry Pi, TI MSPM0, UART

EXPERIENCE

WANG LABORATORY, UT AUSTIN

Austin, TX

Machine Learning Researcher

Feb 2026 - Present

- Built a Python signal-processing pipeline for a 12-subject gut-brain study — automated artifact removal, spectral and phase-amplitude coupling features from EEG/EGG recordings.
- Preparing the resulting feature set for machine learning models that predict optimal timing for targeted ultrasound vagus nerve stimulation.

LONGHORN NEUROTECH

Austin, TX

Edge AI Team Lead

Sep 2025 - Present

- Directed a team of student engineers to optimize neural networks for deployment on embedded systems (Raspberry Pi series).
- Designing a reusable quantization pipeline using ExecuTorch and an inference backend to support keyword spotting and future neuroprosthetic control applications.
- Applied static [INT8 quantization and ONNX](#) export to an MLP classifying hand gestures from 8-channel EMG, reaching 0.43 ms/sample on a Raspberry Pi Zero 2 W against a 1.04 ms FP32 baseline, at a 0.13-point accuracy cost.

AI/ML Developer

Sep 2024 - Aug 2025

- Evaluated CNN and Capsule Network architectures for EEG motor-imagery classification, running Optuna searches over kernel sizes, layer depth, and dropout, and using Captum LayerGradCam for attribution analysis.
- Rearranged EEG channel ordering to match scalp topography so the capsule network's spatial pooling grouped physically adjacent electrodes.

FUNCTIONAL OPTICAL IMAGING LAB, UT AUSTIN

Austin, TX

Researcher and Programmer

Sep 2023 - May 2025

- Developed a real-time [Laser Speckle UI](#) in Python, using Basler cameras and Arduinos for automated laser intensity control.
- Implemented contrast analysis algorithms for laser speckle imaging, improving measurement precision in real-time optical data.

TECHNICAL PROJECTS

STEALTH CLINICAL AI STARTUP (FOUNDER)

Jun 2026 - Present

- Built an [iOS app](#) for a stealth-stage clinical AI startup: it records clinical conversations and analyzes them with a language model entirely on the phone, so no audio or transcript leaves the device.
- Deployed a 4-bit Qwen 3.5-4B on iPhone (llama.cpp, Metal GPU) in under 500 MB of RAM via memory-mapped weights; benchmarked models on 240 labeled decisions, where the 4B beat a 7B (85% vs 79%) with a 30% smaller download.
- Raised agreement with hand-labeled grades from 65% to 84% using a second-pass verifier, and cut per-evaluation latency ~60% (18.6 s → 7.3 s) by reusing the cached transcript KV state.

LOCALABS - ON-DEVICE LAB REPORT TRANSLATOR

Apr 2026 - Present

- Built an [iOS app](#) that translates clinical lab reports into plain language using MedGemma 4B, Google's medical-domain model, running fully on-device via llama.cpp on Metal GPU; no cloud, no account, no telemetry.
- Designed an Apple Vision OCR → LLM pipeline producing five structured sections per report (summary, glossary, diet, questions, medications) in ~12s on iPhone 17 from a 2.5 GB quantized model.
- Contextualized each report with 30 days of HealthKit activity, sleep, and vitals data, interpreting markers against age and sex-bracketed reference ranges from peer-reviewed sources.

CLOUD TO EDGE HEART RATE MONITOR

Dec 2025 - Present

- Architected a cloud-to-edge [pipeline](#) that streams MIT-BIH ECG recordings from Google Cloud Storage to a Raspberry Pi 5 for real-time patient simulation.
- Built a custom C inference engine on the Pico's dual-core Cortex-M0+ that runs a quantized neural network on the incoming stream, classifying each heartbeat as normal, supraventricular ectopic or ventricular ectopic.